YUXIN YANG

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RESEARCH INTERESTS

Large Language Models, Retrieval-Augmented Generation, Test-time Scaling, Graph Machine Learning

EDUCATION

Ph.D. in Computer Science, University of Southern California — Sep 2023 –

- **GPA**: 3.83/4.00
- **Core courses**: Parallel and Distributed Computation (4.0), Advanced Analysis of Algorithms (4.0), Theoretical Machine Learning (4.0), The Science of Large Language Models (4.0)

B.Eng. in Automation, Tsinghua University — Sep 2019 – Jun 2023

- **GPA**: 3.84/4.00

SELECTED AWARDS AND HONORS

- Capital One PhD Fellow — 2024 – 2026
- Changtong Scholarship (top 1% in the Department) — 2022
- Comprehensive Scholarship of University (top 5% in University) — 2020, 2021

PUBLICATIONS AND MANUSCRIPTS

Yuxin Yang, Gangda Deng, Ömer Faruk Akgül, Nima Chitsazan, Yash Govilkar, Akasha Tigalappanavara, Shi-Xiong Zhang, Sambit Sahu, Viktor Prasanna. "SPARC-RAG: Adaptive Sequential-Parallel Scaling with Context Management for Retrieval-Augmented Generation." Preprint, 2026.

Abdulla Alshabanah, **Yuxin Yang**, Murali Annavaram. "SAGERec: Sampling and Gating for Enhanced Long-Tail Item Recommendations." 18th ACM International Conference on Web Search and Data Mining (WSDM), 2026.

Gangda Deng*, **Yuxin Yang***, Ömer Faruk Akgül*, Hanqing Zeng, Yinglong Xia, Rajgopal Kannan, Viktor Prasanna. "Training Diverse Graph Experts for Ensembles: A Systematic Empirical Study." Preprint, 2025.

Ömer Faruk Akgül*, Feiyu Zhu*, **Yuxin Yang**, Rajgopal Kannan, Viktor Prasanna. "RECIPE-TKG: From Sparse History to Structured Reasoning for LLM-based Temporal Knowledge Graph Completion." Preprint, 2025.

Yuxin Yang, Hongkuan Zhou, Rajgopal Kannan, Viktor Prasanna. "Towards Ideal Temporal Graph Neural Networks: Evaluations and Conclusions after 10,000 GPU Hours." 51st International Conference on Very Large Data Bases (VLDB), 2025.

Yuxin Yang, Yitao Liang, and Muhan Zhang. "PA-GNN: Parameter-Adaptive Graph Neural Networks." Workshop on Dynamic Neural Networks in *the 39th International Conference on Machine Learning* (ICML-DyNN-Workshop), 2022, oral presentation.

RESEARCH EXPERIENCE

SPARC-RAG: Adaptive Sequential-Parallel Scaling with Context Management for RAG — Jan 2025 – Jan 2026

Research Assistant, advisor: Prof. Viktor Prasanna — University of Southern California

In collaboration with Capital One Research

- Proposed **SPARC-RAG**, a multi-agent framework that adaptively scales **sequential depth** and **parallel width** for multi-hop Retrieval Augmented Generation (RAG), addressing context contamination and scaling inefficiency in naive scaling approaches.

- Designed a **unified context management** mechanism with specialized agents maintaining shared global context, enabling explicit control over the scaling process and avoiding context pollution.
- Implemented **dynamic query generation** that produces targeted, complementary sub-queries for each branch, enabling diverse parallel exploration of the reasoning space.
- Developed **lightweight fine-tuning** with process-level verifiable preferences for adaptive exit decisions based on answer correctness and evidence grounding.
- Achieved **+6.2 average F1 improvement** on multi-hop QA benchmarks (HotpotQA, 2WikiMultiHopQA, MuSiQue, Bamboogle) under lower inference cost.

Towards Ideal Temporal Graph Neural Networks: Evaluations and Conclusions Oct 2023 – May 2024
Research Assistant, advisor: Prof. Viktor Prasanna University of Southern California

- Designed a unified and fully optimized TGNN comparison framework that isolates and evaluates core modules (neighbor sampling, node memory, aggregation), enabling fair design-space exploration across models.
- Conducted a detailed experimental study to answer three key research questions on efficiency, universality, and module interplay; identified that most-recent sampling, attention aggregation, and node memory yield optimal cost-performance tradeoffs.
- Discovered that dataset repetition patterns fundamentally determine node-memory preference, showing RNN-based memory excels on short-term repetition datasets while static embedding memory excels on long-term repetition datasets.
- Revealed strong diminishing returns in deeper/wider neighbor sampling when node memory is present, leading to practical design recommendations for future TGNN architectures.

PA-GNN: Parameter-adaptive Graph Neural Networks Nov 2021 – Jun 2022
Research Assistant, advisor: Prof. Muhan Zhang Beijing Institute for General Artificial Intelligence

- Showed that capturing local patterns of graph datasets improves performance of learning tasks on graphs.
- Designed a trainable node-specific aggregator that learns from node position and features to capture local patterns; Utilized DeepWalk to encode node position.
- Experimentally showed that graph neural networks with an adaptive aggregator can exploit local patterns and improve node classification accuracy by up to 5%.

COMPUTER AND LANGUAGE SKILLS

- **Programming**: Python, C/C++, Javascript
- **Tools**: PyTorch, MATLAB, CMake, LabVIEW, Linux, Git, LaTeX
- **Language**: TOEFL 114/120 (Reading 30, Listening 29, Speaking 25, Writing 30); GRE 327/340+3.5/6.0 (Verbal 157, Quantitative 170, Analytical Writing 3.5)